Kaustubh Hemant Sonawane

Bristol, UK | ksstubh.sonawane@gmail.com | +44 7388229743 | +91 8591441019 | www.github.com/Arishtanemi4 | www.linkedin.com/in/kaustubhson4 | https://kaustubh-sonawane-portfolio.vercel.app

UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Machine Learning and AI Engineer with 3 years of industry experience owning production ML and GenAI systems end to end - technical scoping, architecture, software, deployment and production troubleshooting - in a regulated financial services setting. Builds production-grade ML software and scalable cloud infrastructure with **Python**, **Docker** and **AWS**, including architecture, security (**RBAC**, **LDAP**, **VAPT**) and a one-click **Jenkins** and **GitLab CI/CD** pipeline, and turns business requirements into shipped systems. Designed the **scoring and ranking algorithm** behind an **AstraZeneca MSc dissertation** that ranks cancer cell lines by target inclusion and exclusion genes, and completed a multi-architecture deep learning study. Led up to 4 interns as a **de facto team lead** and was twice recognised with **Best Team Project** (2024, 2025). Seeking to bring that ownership to **Faculty's Life Sciences team**.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Production ML Software & GenAI Systems

- Designed and built production ML and GenAI applications end to end (**Python FastAPI** microservices, a **ReactJS** frontend, retrieval with **LlamaIndex** and **Azure OpenAI GPT-3.5 & Llama-3**) for the Compliance, Client Service and Investment teams, with a separate **pgvector** vector database and **Azure OpenAI embeddings** per department.
- Owned the model and cost decisions as engineering trade-offs: self-hosted **Llama-3 on a GPU EC2 instance** for the largest department and kept the stronger **Azure OpenAI** outputs for decision makers; built **agentic tool-calling** where agents call **Python functions** to run **SQL-backed fund comparisons** and generate **PDF reports** on request.
- Took the systems from prototype to real users and iterated on what broke, guardrailing behaviour through **system prompts** (no fine-tuning), versioning every prompt change in **Git** and routing each release through **Compliance review**.

Cloud Architecture, Security & Deployment

- Designed the infrastructure architecture for the analytics platform and GenAI applications on **AWS** (**EC2** including a GPU instance, **S3, ECR and Lambda**); packaged the FastAPI microservices as **Docker** containers and automated deployment of the separate frontend and backend repositories (dev and prod) through a **one-click Jenkins and GitLab CI/CD pipeline**.
- Worked with the security team on the fixes: **RBAC** via a Python function against the company **LDAP** server, access restricted to company private IPs with **AWS Security Groups**, and an **Nginx reverse proxy**, one of the fixes that took the applications from repeatedly failing to passing the **VAPT**.
- Ran real-time and batch workloads side by side: user-facing services on FastAPI microservices, plus a daily rating-email pipeline on **AWS SageMaker** scheduled by an **EventBridge** rule that triggers a **Lambda** function at 9 AM; troubleshot and fixed it in production when a rating agency changed its email format.

Reusable Systems, Data & Cross-Functional Delivery

- Built **reusable components shared across the department applications** (SQL-backed tool functions, an ingestion and embedding pipeline, the FastAPI microservice set and an LDAP authentication function), working alongside a senior engineer and a dedicated frontend developer as the team grew.
- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025); built a **Python extraction pipeline** that parses over **300 daily emails** from credit rating agencies, saving **60 analyst hours a month**.
- Worked directly with **Client Service, Compliance and Investment** stakeholders to scope requirements, turn them into working systems and explain the architectural trade-offs; **led up to 4 interns as de facto team lead**; recognised with **Best Team Project** twice (2024, 2025).

PROJECTS

- University of Bristol MSc. Data Science Dissertation in partnership with **AstraZeneca**: CellLineSelector (**Team project, co-built**; I designed the **scoring and ranking algorithm** that ranks roughly **2,100 cancer cell lines** across six evidence layers by target inclusion and exclusion genes, the **data preprocessing pipeline** over DepMap, GEO and HPA data and the **RAG-based AI assistant**; FastAPI microservices and a React/TypeScript frontend; validated with feedback from **AstraZeneca scientists**.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- University of Bristol MSc Applied Data Science Group Project: VisionCaptioner (**Team project.** Multi-architecture deep learning study comparing **CNN/ViT/ResNet encoders** with **LSTM/attention decoders** for generating natural-language image descriptions, benchmarked against BLIP/SigLIP and commercial VLMs.) GitHub: https://github.com/Arishtanemi4/ata-team30.git
- Wingman: AI dating-coach web app with **108 LLM personas** across 12 archetypes and a **vision model** that scores uploaded photos (ReactJS, Python FastAPI, NVIDIA NIM). LLM output treated as untrusted: validated with **Pydantic**, repaired and cached by input hash; rate limits shaped the design (12 calls instead of 108, results streamed to the UI). GitHub: https://github.com/Arishtanemi4/wingman.git
- B.Tech Final Year Project: Udacity self-driving car simulator. (A self-driving car trained using a **Convolutional Neural Network (CNN)** model.)

ACHIEVEMENTS

- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- **Tools:** Jupyter, Anaconda, AWS (EC2 incl. GPU, S3, ECR, Lambda, SageMaker, EventBridge), Azure OpenAI, LlamaIndex, Docker, Jenkins, GitLab, Git, Nginx, Linux, LDAP, Power BI, VS Code, Claude Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI, Pydantic, uvicorn), SQL, R (dplyr, tidyverse, ggplot2), NodeJS (express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector, SQL functions and procedures), SQLite